

SD311 AML-ML

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What you will be evaluated on (i.e. what you will learn)

Technical skills :

- Hands on practice of all major algorithms (with **sklearn** and **keras**)
- Hands on practice of data analysis tools (**Jupyter**, **bokeh/plotly**, **pandas**)
- Key principles of all major algorithms
- Main bottlenecks of data driven approaches

Methodology skills :

- Use the correct vocabulary from the field
- Choose the correct class of algorithm for each problem
- General Knowledge of the history of the field
- Present the results to aid decision

Planning of the module

2 Oct.	8 Oct	15 Oct	5 Nov.
8h30 - 11h45 : Vocabulary [0] Data Analysis [1,2]	8h30 - 11h45 Bayes, Regression and Gaussian processes [5,6]	8h30 - 11h45 : Ensemble method Boosting [8,9]	9h - 11h Explainability [14]
	9 Oct	22 & 23 Oct	6 Nov.
13h30 - 16h45 : Supervised learning with SVM [3,4]	8h30 - 11h45 Surrogate Modelling, Bayesian optim [7]	8h30 - 11h45 XGboost practice [10] Bagging & Random forest [11,12]	8h30 - 11h45 : Anomaly detection [13+ evaluation]

+ 5 optional home exercices



Links

Courses notebooks :

<https://supaerodatascience.github.io/>

<https://github.com/SupaeroDataScience/machine-learning/>

<http://scikit-learn.org>

<https://datasetsearch.research.google.com/>

<https://www.kaggle.com/>

<https://www.datascienceweekly.org/>

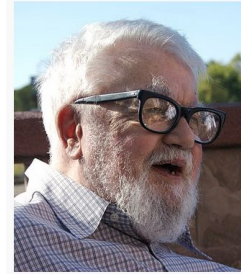
A definition of AI ?



WIKIPÉDIA
L'encyclopédie libre

Artificial intelligence (AI) is a “set of theories and techniques implemented to create machines capable of simulating human intelligence”

“The construction of computer programs that perform tasks that are, for the moment, accomplished more satisfactorily by human beings”



*John McCarthy
AI Pioneers with M.L. Minsky*

Programs that solve complicated tasks: those that are only accomplished today by humans

The different types of learning

Supervised Learning

- Learning with a **labeled** training set.

*Learn with exercises
Ex. Driving license*

Unsupervised Learning

- Discovering patterns in **unlabeled** data.

*Learn with similitude
Ex. Newton and the apple*

Reinforcement Learning

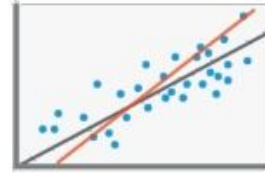
- Learning based on **feedback** or **reward**.

*Learn with trial and error
Ex. Ride a bike*

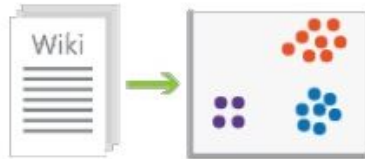
ML to solve different types of problems



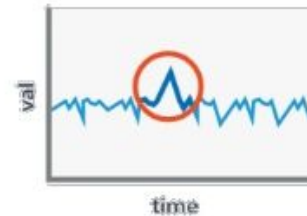
Classification
(supervised – predictive)



Regression
(supervised – predictive)



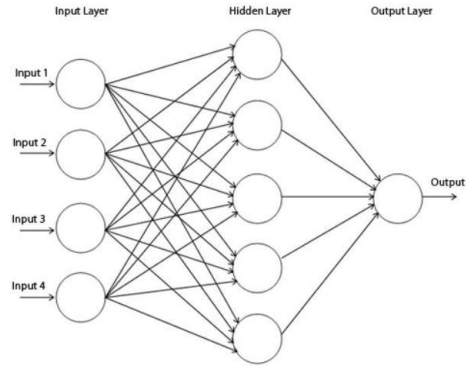
Clustering
(unsupervised – descriptive)



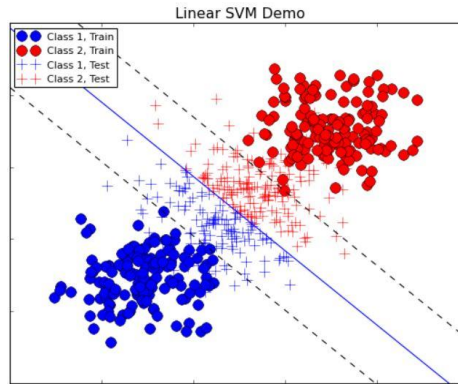
Anomaly Detection
(unsupervised – descriptive)

Classical Machine Learning

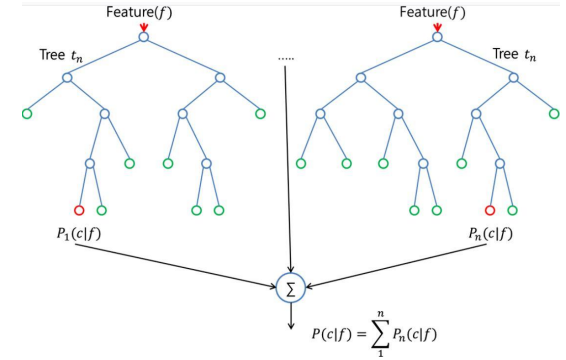
Multi-Layer Perceptron (1986)



SVM (1995)



Random forest (2001)



A brief history of Deep Learning

1950

- Test de Turing

1981

- Fukushima Neocognitron : lecture d'écriture manuscrite en Japonais

1988:

- Convolutional Network (**CNN**) de LeCun
lecture d'adresse postale. 60k paramètres

2012

- Traffic Signs Challenge : Performances meilleures que les humains. AlexNet : 60 M paramètres

2016

- Alphago bat le champion du monde de go.

2024

- GPT4o : 8*220 Milliards de paramètres

Google Gemini : 1560 Milliards de paramètres



MIT
Technology
Review

Facebook Launches Advanced AI Effort to Find Meaning in Your Posts

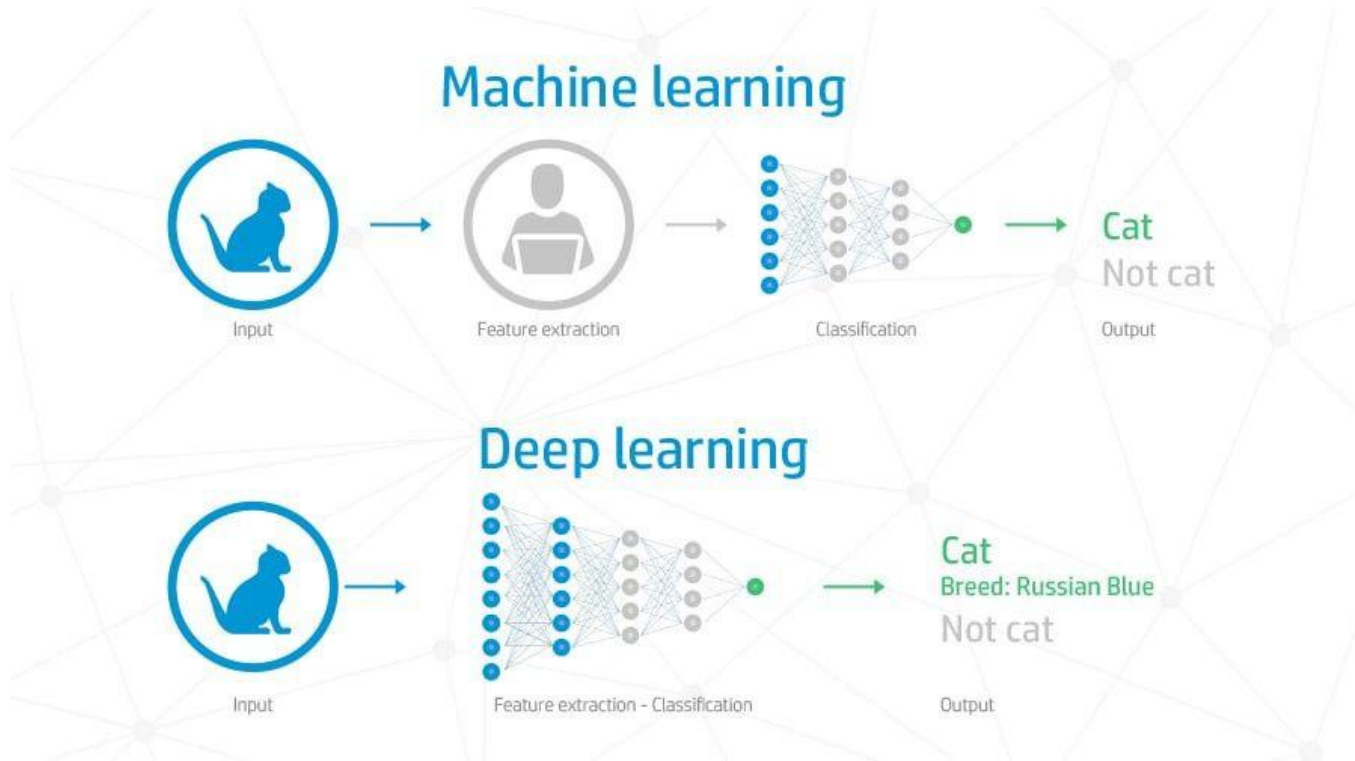
A technique called deep learning could help Facebook understand its users and their data better.



© reuters/ Kim Hong Ji



Machine Learning != Deep Learning != Artificial Intelligence



Solved Applications



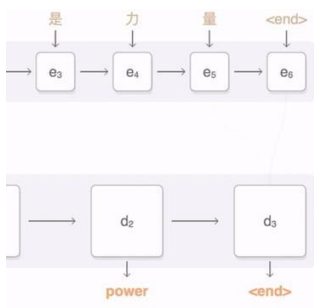
Image Classification :
92% on Image Net



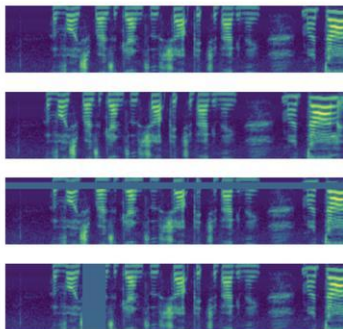
Object Detection



Sentiment analysis
(amazon, twitter, ...)
96% on IMDB



Machine Translation
BLEU score 40
(34 human pro)



Speech Recognition
97% on Noisy

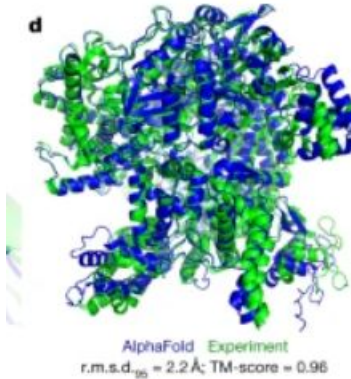


Atari, Chess, Go

Applications still under research



Image & Video Generation
Diffusion Models



Protein Prediction
>90% AlphaFold 2



Conversation agents (LLM)
67,6% Accuracy on US Medical
exam



Multi agents games :
Starcraft, Diplomacy...

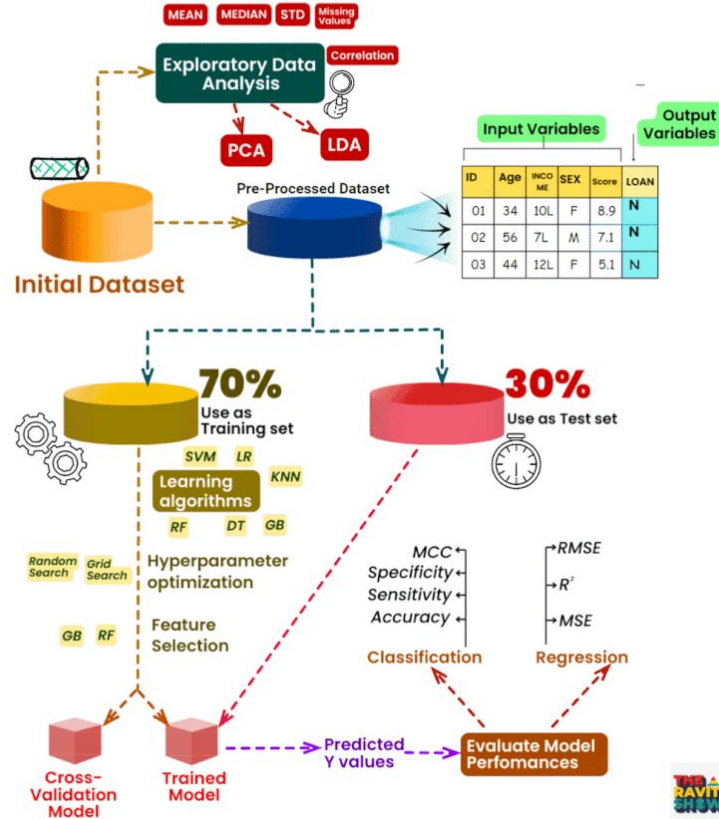
Exercice 1 : Regression

Objectives :

- Understand the difference between Regression and Classification
- Understand the definition of a Label

Working of Machine Learning Model

Don't Forget to
Save For Later



Evaluation criterias

Evaluating supervised ML methods: what do we really want?

Ability to fit the training data (regression):

- Mean Square Error, coefficient of determination.

$$MSE = \frac{1}{N} \sum_i (y_i - f(x_i))^2$$

$$R^2 = 1 - \frac{\sum_i (y_i - f(x_i))^2}{\sum_i (y_i - \bar{y})^2}$$

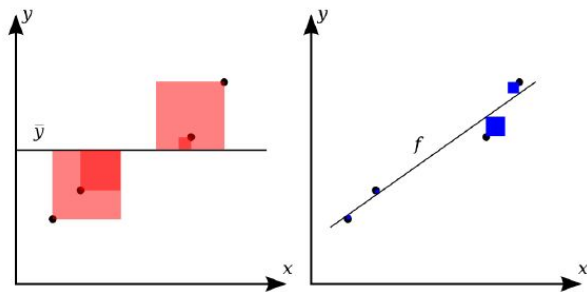


Image source: Wikimedia commons

Evaluation criterias

Ability to fit the training data (classification):

- Accuracy, TP, FP, confusion matrix [link]

$$Acc = \frac{TP + TN}{TP + FP + TN + FN}$$

- ROC, AUC, [link]

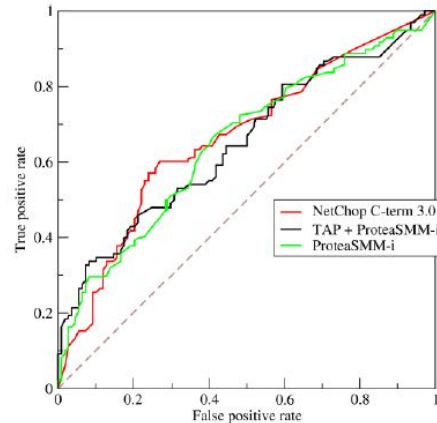
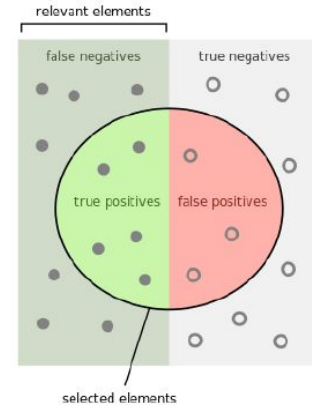


Image source: Wikimedia commons



How many selected items are relevant?

Precision = $\frac{\text{true positives}}{\text{true positives} + \text{false positives}}$

How many relevant items are selected?

Recall = $\frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$

Image source: Wikimedia commons

Evaluation criterias

Ability to generalize:

- Goal: filter out noise, avoid overfitting, generalize to unseen cases.
- ML Notions:
 - maximize margin
 - minimize difference btw class distributions (cross-entropy [\[link\]](#))

$$H(p, \hat{p}) = \sum_i p(x_i) \log(\hat{p}(x_i)) = \mathbb{E}_p(\log(\hat{p}))$$

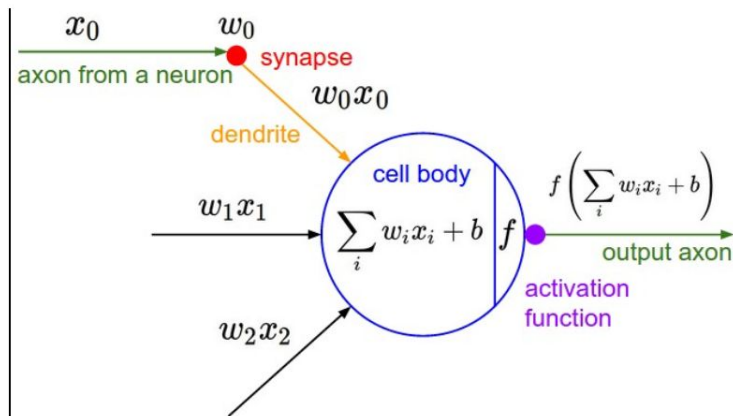
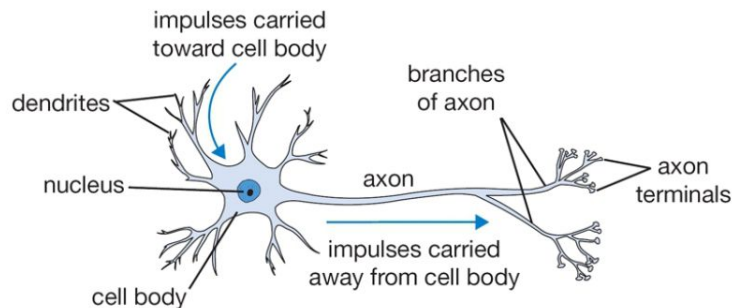
Exercice 2 : Features

Objectives :

- Understand the notion of Feature
- Understand the importance of Feature selection
- Understand how Deep learning changes the computation of Features

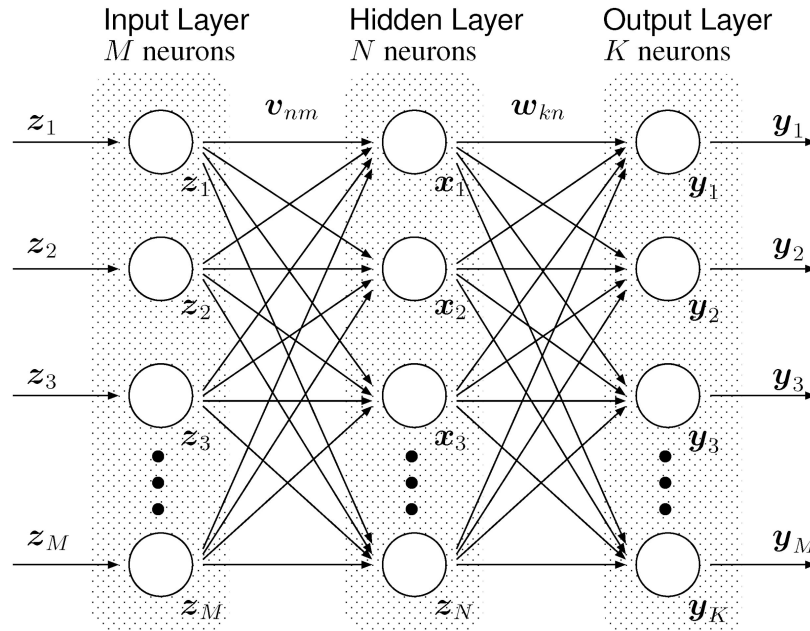
Neurons

- Neurons are trained to filter and detect features such as edges, shapes, textures, by receiving weighted inputs from the previous neurons, transforming it with an activation function and passing it to the outgoing connections.



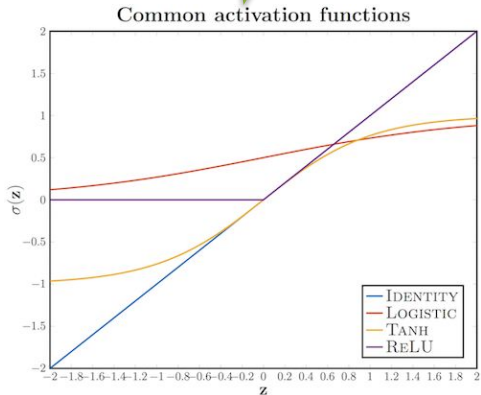
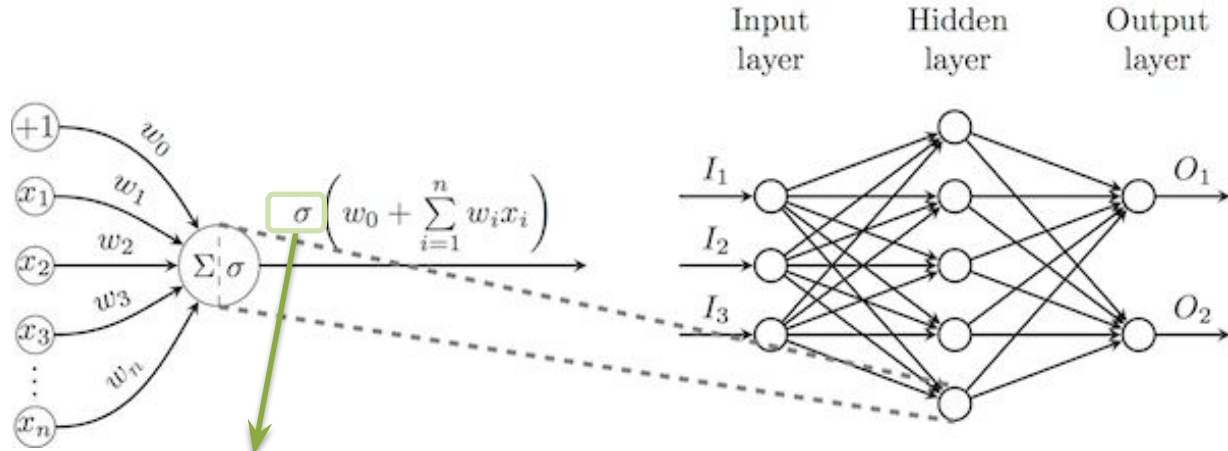
Multi-layer Perceptron (MLP)

- MLP interest is in the association of neurons in multi layers : it results in a composition of non linear functions that can represent complex problematics.



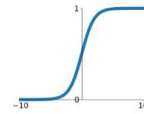
- Parameters estimation:
- Quadratic error is known (estimated – known)² => we can estimate the gradient for the last layer
- We don't know the quadratic error associated to each hidden layer.

Activation Functions



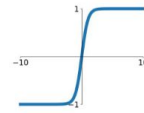
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



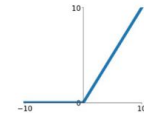
tanh

$$\tanh(x)$$



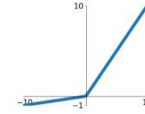
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

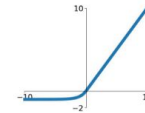


Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$



Exercise 3 : Neurones

Objectives :

- Understand the influence of hyper-parameters
- Reinforce the notion of Feature and the distinction between ML and DL

■ playground.tensorflow.org/

Learning contexts

Different kinds of learning contexts:

Context	Sample source
▶ Offline, batch, non-interactive	all samples are given at once
▶ Online, incremental	samples arrive one after the other
▶ Active	the alg. asks for the next sample

Quizz time : Fill in the definitions

<i>Level 1</i>	Machine Learning Apprentissage avec des données	Deep Learning ML avec des réseaux de neurones	Artificial Intelligence Algorithmes qui réalisent des tâches considérées comme humaines	Big Data Grandes bases de données
<i>Level 2</i>	Supervised vs Unsupervised learning Supervisées: avec label Non supervisés: sans label	Classification vs Regression Classification: mettre les données dans des classes Régression: prédiction de valeurs	Correlation Existence de liens entre les features, ou entre features et targets	Feature vs target Entrées et sorties
<i>Level 3</i> <i>[you are here]</i>	Overfitting Lorsque le modèle a trop bien appris avec les données d'entraînement et ne généralise plus sur les données de test	Hyper parameter Paramètres qui ne sont pas les paramètres d'entraînement mais des paramètres intrinsèques à l'apprentissage du modèle: learning rate, fonction d'activation, etc	Training vs Testing Dataset Séparation du data set	Feature engineering Etape avant l'apprentissage pour transformer les features, les caractéristiques pour les trier etc
<i>Level 4</i>	ROC curve Courbe du taux de vrais positifs sur le taux de faux positifs	Cross validation On entraîne et on teste plusieurs fois sur différentes parties du data-set Ex: K-folds	Gradient descent Résolution de problème d'optimisation	Bias vs Variance Compromis à trouver entre biais et variance. Biais: à quel point ce qu'on dit est vrai Variance: sensibilité aux données (d'entrée, d'entraînement, ...)

Quizz time : Some answers

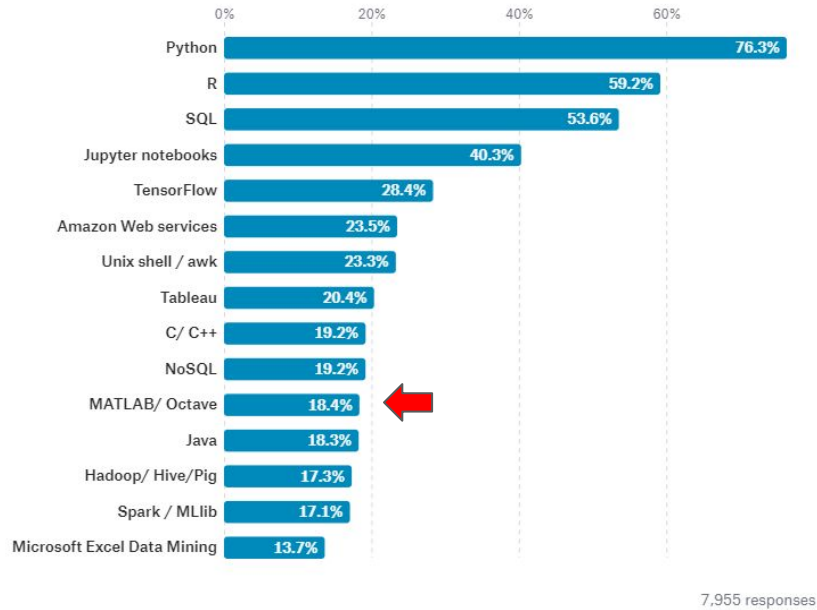
Machine learning is a field of computer science that gives computer systems the ability to “learn” (i.e. progressively improve performance on a specific task) with data, without being explicitly programmed. (Wikipedia)

Artificial intelligence (AI) is the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings. (Brittanica)

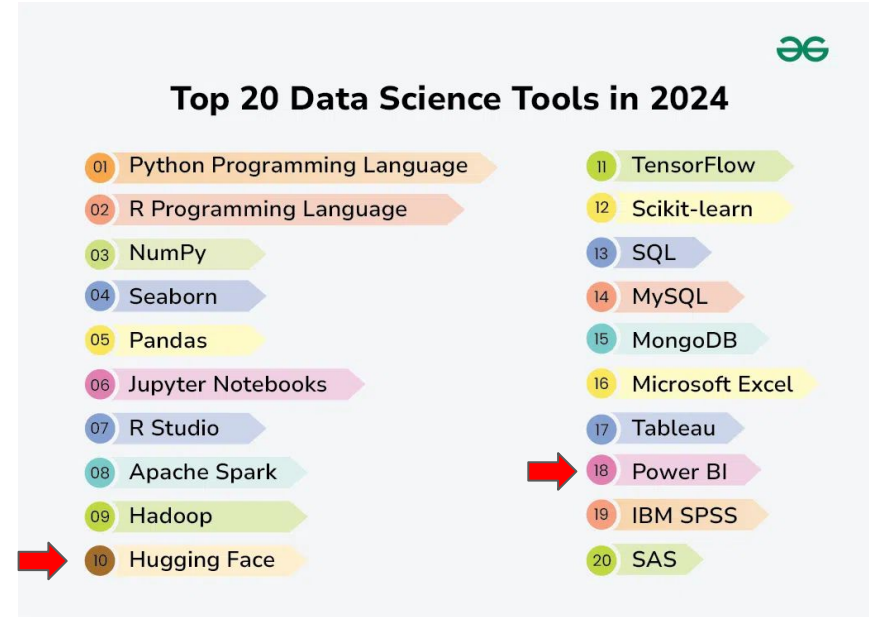
Big Data refers to working with datasets that have large Volume, Variety, Velocity (, Veracity, and Value).

Deep Learning is Machine Learning with Deep Neural Networks.

Which tools are used



2017



2024

... and a lot of tools to put ML/DL in production

Cloud is the easiest option (by far...)
Optimized stacks for training and inference

But ...

Exploding costs depending on the number of parameters

QUICK START WITH CLOUD PARTNERS

Get up and running with PyTorch quickly through popular cloud platforms and machine learning services.



Amazon Web Services >



Google Cloud
Platform >



Microsoft Azure >

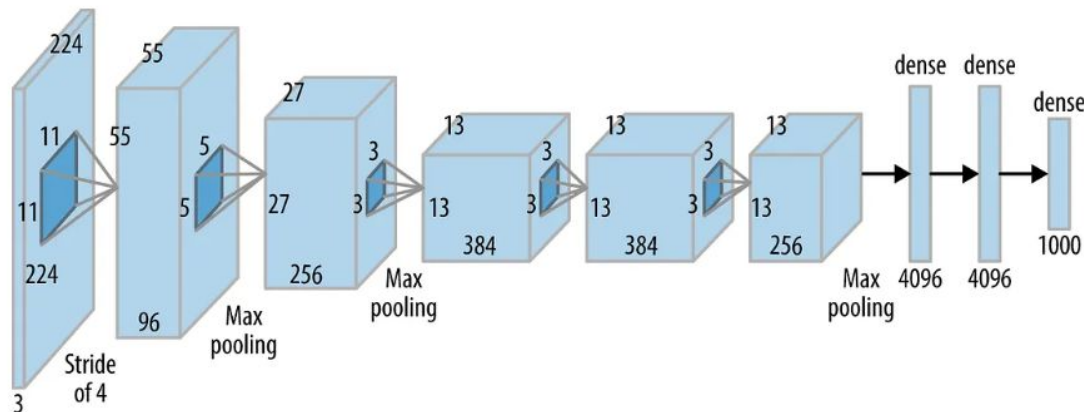
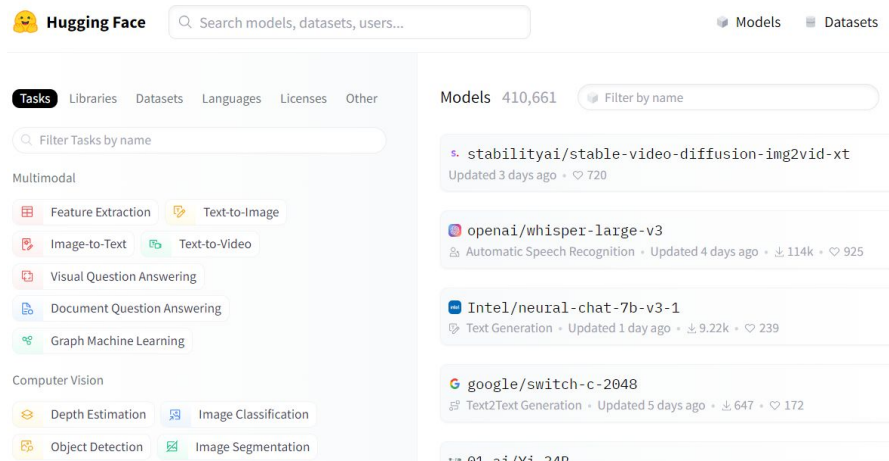
<https://pytorch.org/>

	Training cost
DeepMind AlphaGO	35 Million \$
GPT3	12 Million \$
<i>CoAtNet (top 1 ImageNet)</i>	250 000 \$
BERT	7000 \$
Yolo V5	100 \$
ResNet 50	10 \$

Start from a model already trained

Hugging Face / Pytorch hub :
state of the art models with
weight already tuned
=> we add images, continue
training and voilà !

Articles and blogs
describe
architectures (how
many layers, which
types), which are
known to work well
on a given problem



What should I look for in a data scientist's CV?

Must have :

- Technology names (most of them) : sklearn, python / R, pytorch / keras / tensorflow, jupyter, numpy, pandas, spark
- Experiences with datasets outside of a MOOC
- Likes understanding people's problems

Nice to have :

- PhD (in computer science, applied math or physics)
- kaggle competition/score
- publications (Arxiv, JMLR, MLJ, IEEE PAMI, NIPS, ICML, ICLR...)
- cloud experience (AWS, GCP, Azure) or deployment experience (docker, terraform, kubernetes,...)

Sklearn : let's have a look

<http://scikit-learn.org>

The screenshot shows the scikit-learn website's User Guide page. The header includes the scikit-learn logo and navigation links: Install, User Guide, API, Examples, and More. A search bar with a 'Go' button is on the right. The left sidebar contains navigation buttons (Prev, Up, Next), the current version (scikit-learn 1.0.2), a citation notice, and a list of sections under 'User Guide'. The main content area displays the 'User Guide' title and a list of supervised learning topics.

scikit-learn

Install User Guide API Examples More ▾

Prev Up Next

scikit-learn 1.0.2
Other versions

Please [cite us](#) if you use the software.

User Guide

1. Supervised learning
2. Unsupervised learning
3. Model selection and evaluation
4. Inspection
5. Visualizations
6. Dataset transformations
7. Dataset loading utilities
8. Computing with scikit-learn
9. Model persistence
10. Common pitfalls and recommended practices

User Guide

1. Supervised learning

- ▶ 1.1. Linear Models
- ▶ 1.2. Linear and Quadratic Discriminant Analysis
- 1.3. Kernel ridge regression
- ▶ 1.4. Support Vector Machines
- ▶ 1.5. Stochastic Gradient Descent
- ▶ 1.6. Nearest Neighbors
- ▶ 1.7. Gaussian Processes
- ▶ 1.8. Cross decomposition
- ▶ 1.9. Naive Bayes
- ▶ 1.10. Decision Trees
- ▶ 1.11. Ensemble methods
- ▶ 1.12. Multiclass and multioutput algorithms
- ▶ 1.13. Feature selection

Toggle Menu